Machine Learning and its Applications

Today's goals

Module 1
Machine Learning Fundamentals I: Basic Concepts

Welcome

Welcome to the specialization - Applied AI for Engineers and Scientists: Foundations
Video • 6 min

Specialization and Course Structure
Reading • 10 min

Specialization Sample Certificate
Reading • 10 min

Stay connected with UofG Online
Reading • 5 min

Module 7 Introduction

Introduction to Module 7's Study
Video • 1 min

How to Access MATLAB Online
Reading • 5 min

Machine Learning in Engineering Practice: Tool, Replacement, or Decision Aid?
Reading • 5 min

Module 7-1 Machine Learning Fundamentals

Machine Learning Fundamentals: What is Machine Learning
Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (1)
Video • 6 min

Machine Learning Fundamentals: Fundamental Concepts in Machine Learning (2)
Video • 8 min

Materials for Machine Learning Fundamentals
Reading • 10 min

From Explicit Rules to Learned Models: What Actually Changes?
Reading • 5 min

Module 1 Learning Coach
Dialogue • 5 min

Module 7-2 Mapping Inputs to Outputs

Mapping Inputs to Outputs: Data Representation
Video • 10 min

Mapping Inputs to Outputs: Parametric ML Model
Video • 7 min

Mapping Inputs to Outputs: Non-Parametric ML Model
Video • 7 min

Mapping Inputs to Outputs: Evaluate Output
Video • 2 min

Materials for Mapping Inputs to Outputs
Reading • 10 min

How Data Representation and Model Choice Shape What a Model Can Learn
Reading • 5 min

Module 7-3 MATLAB Implementation

MATLAB Implementation: Simple Linear Regression and KNN
Video • 6 min

Materials about MATLAB Implementation
Reading • 10 min

From Concept to Code: What the Model Is Actually Doing
Reading • 5 min

Module 7 Quiz

Module 7 Quiz
Practice Assignment • Grade: 100%

Module 7 Assignments

Module 7 Assignment 1
Graded App Item • 30 min

Module 7 Assignment 2
Graded App Item • 30 min

CSV File for Module 7 Assignment
Reading • 1 min

Module 7 Recap
Reading • 5 min

Module 2
Machine Learning Fundamentals II: Model Training and Evaluation

Module 3
Data Preparation

Module 4
Support Vector Machines

Module 5
Artificial Neural Networks

Module 7 Recap

0:05 / 2:06 2x

From Data to Decisions: Reflecting on the Machine Learning Workflow

This module, you were introduced to machine learning not as a collection of algorithms, but as a **process** for mapping inputs to outputs in science and engineering problems.

1. Framing the problem

We began by distinguishing machine learning from traditional programming and by identifying when ML is appropriate.

Think of a **real engineering or scientific problem** you care about.

- What is the input data?
- What is the output you want to predict?
- Is this naturally a classification or regression problem, and why?

What assumptions are you *implicitly* making by choosing this formulation?

2. Representing data

We explored how raw data must be represented before a model can learn.

- What types of data does your problem involve (numerical, categorical, ordinal, nominal)?
- Which representation or encoding choices feel *natural* to you, and which feel forced?
- How might a poor representation limit what *any* model can learn, no matter how advanced?

3. Choosing a model

We compared parametric and non-parametric models using linear regression and KNN.

- Which type of model do you instinctively trust more: a simple, interpretable model or a flexible, data-driven one?
- What trade-off are you making when you choose simplicity over flexibility (or vice versa)?
- Can you think of a case where a model gives “reasonable” predictions for the *wrong reasons*?

4. Evaluating outcomes

We briefly introduced prediction error as a way to judge model quality.

- What does a prediction error actually tell you about your model?
- When might a low error be misleading?
- How confident would you feel deploying a model without understanding *why* it performs well or poorly?

5. Looking ahead

In the next module, we move from building models to **training and evaluating them properly**.

What questions do you now have about:

- training vs test data,
- overfitting and underfitting, or
- trusting model performance?

What do you feel least confident about as we transition into model training and evaluation?

Mark as completed

Like Dislike Report an issue

Go to next item →